Bengaluru - 560059, Karnataka, India

Department of Artificial Intelligence and Machine Learning

Date: 10.011.2025	CIE 1	Max. Marks: 50+10
Semester: V	UG	Duration: $1\frac{1}{2}$ Hrs + ½ Hr
Title: Mathematical Algorithms for Artificial Intelligence		Course Code: AI255BTC

Part A

Q. No		PART A Quiz 1	M	BT	CO
1	a)	.**Statement 1**: A system of linear equations is consistent if it has a unique solution. **Statement 2:** A system of linear equations is inconsistent if it has infinitely many solutions. Identify the correct option with respect to a consistent system. (a) **ONLY** Statement 1 is **TRUE** (b) **ONLY** Statement 2 is **TRUE** (c) Both Statement 1 and Statement 2 are TRUE (d) Both Statement 1 and Statement 2 are FALSE	2	1	1
	b)	Which among the following is **NOT** an algebraic field? (a) R_{57} the set of reminders modulo 57 (b) R_{53}, the set of reminders modulo 53 (c) R_{47}, the set of reminders modulo 47 (d) R_{61}, the set of reminders modulo 61 (e) All of them are fields.	2	1	2
	c)	Which among the following is not a vector space over $\mathbb{R}$, the field of real numbers, with real number addition and multiplication operations (a) $\mathbb{R}^2$ (b) $S_1 = \left\{\begin{pmatrix}2\\3\end{pmatrix}\right\}$ (c) $S_2 = \left\{\begin{pmatrix}x_1\\x_2\end{pmatrix} : x_2 = 0\right\}$ (d) $S_3 = \left\{\begin{pmatrix}x_1\\x_2\end{pmatrix} : x_2 = 4x_1\right\}$	2	2	1
	d)	You are given the following statements: **Statement 1: A vector space is a linearly dependent set** **Statement 2: A basis for a vector space is a linearly independent set.** (a) Both statements are FALSE (b) Both statements are TRUE (c) Statement 1 is ALWAYS FALSE, STATEMENT 2 IS ALWAYS TRUE (d) Statement 1 is ALWAYS TRUE, STATEMENT 2 is ALWAYS FALSE	2	2	2
	e)	Which of the following is NOT TRUE about a set containing only one non-zero vector? (a) The set is a linearly independent set (b) The set is a basis for a 1-D subspace (c) The set is a vector space over a field of real numbers (d) The span of this set is a line passing through the origin	2	2	2

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<table>
<tr><th>Q. No</th><th>PART B CIE</th><th>M</th><th>B</th><th>CO</th></tr>
<tr><td>1</td><td>Let V be a real vector space. Suppose that $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4$ are vectors in V which are linearly independent. Prove or Disprove the vectors $\mathbf{v}_1, \mathbf{v}_1+\mathbf{v}_2, \mathbf{v}_1+\mathbf{v}_2+\mathbf{v}_3, \mathbf{v}_1+\mathbf{v}_2+\mathbf{v}_3+\mathbf{v}_3$ 4 are also linearly independent.</td><td>6</td><td>2</td><td>1</td></tr>
<tr><td>2</td><td>Let
$$\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}$$
have non-zero solution $\mathbf{x}_h$ to the corresponding homogeneous system of equations and $\mathbf{x}_p$ be a particular solution for the given **b** (**b** $\neq$ **the zero vector). Prove or disprove that the complete set of solutions** $\{\mathbf{x}_{complete} = \mathbf{x}_p + \alpha\mathbf{x}_h\}$ is a vector subspace of $\mathbb{R}^2$, for real scalars α.</td><td>6</td><td>2</td><td>1</td></tr>
<tr><td>3</td><td>Assert or Reject each of the following 3 statements with proper justification.
• **Statement 1:** The set containing a single vector is **ALWAYS** a linearly **DEPENDENT** set.
• **Statement 2:** The set containing a single vector is **ALWAYS** a basis for some vector subspace.
• **Statement 3:** Every vector space contains at least one subspace.</td><td>6</td><td>3</td><td>2</td></tr>
<tr><td>4</td><td>Determine if the following statements are **TRUE** or **FALSE. Justify your answer with an example or a counterexample.**
(a) Every non-zero subspace of say $\mathbb{R}^5$ has infinitely many bases.
• (b) The set of columns of a 3x3 matrix will form a basis for $\mathbb{R}^3$.
• (c) The rank of a matrix A is the number of non-zero rows in it.
• (d) The vector space $\mathbb{R}^{27}$ contains 27 subspaces.</td><td>12</td><td>3</td><td>2</td></tr>
<tr><td>5</td><td>Find the condition under which the following system of equations is consistent.
$$\begin{aligned} 3x_1 + x_2 + x_3 &= b_1 \\ x_1 - x_2 + 2x_3 &= b_2 \\ 2x_1 - 2x_2 + 4x_3 &= b_3 \end{aligned}$$</td><td>10</td><td>3</td><td>2</td></tr>
<tr><td>6</td><td>Let the set $\mathbb{R}^3$ with operations $\oplus$ and $\odot$ defined as follows. For any two vectors $\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}, \mathbf{y} = \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix}$, in the set,
<table><tr><td>vector addition is defined as
$$\mathbf{x} \oplus \mathbf{y} = \begin{pmatrix} x_1 + y_1 + 5 \\ x_2 + y_2 - 7 \\ x_3 + y_3 + 1 \end{pmatrix}$$</td><td>For a real scalar k, scalar multiplication is defined as
$$k \odot \mathbf{x} = \begin{pmatrix} kx_1 + 5(k-1) \\ kx_2 - 7(k-1) \\ kx_3 + k - 1 \end{pmatrix}$$</td></tr></table>
Find the zero vector for this vector space with the operations as defined above.</td><td>10</td><td>3</td><td>2</td></tr>
</table>

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Academic Year: 2025-2026

Department of Artificial Intelligence and Machine Learning

Date: 17/12/2025	**Test – II**	**Max. Marks : 10 + 50**
Semester: 5^{th}	**UG**	**Duration : 2 hours**
Course Title: Mathematical Algorithm for AI		**Course Code: AI255TBC**

Part – A

QN	Questions	M	BT	CO
1.1	If for two vectors u, v, u.v = 0, then (a) u and v are collinear (b) u and v are orthogonal (c) u and v are linearly dependent (d) u and v have no relation	2	2	2
1.2	If the characteristic equation of a 3x3 matrix A is $\lambda(\lambda^2 - 4) = 0$ then the determinant of A is (a)0 (b) 4 (c) - 4 (d) indeterminate.	2	3	3
1.3	The orthogonal complement subspace of $\mathbb{R}^2$ is (a) the z-axis (b) the cross product of the basis vectors for $\mathbb{R}^2$ (b) (c) $\left\{\begin{pmatrix}0\\0\end{pmatrix}\right\}$ (d) indeterminate	2	3	2
1.4	Which of the following is not a property of dot product? Here, a, b, c are vectors and k a real scalar. (a) a.b = b.a (b) a.(b+c) = a.b + a.c (c) a.(kb + c) = k(a.b) + a.c d) a.(b.c) = (a.b).c	2	2	3
1.5	If u, v, and w are orthonormal vectors, then $3u^Tv - 2v^Tv + 3u^Tw + 3w^Tw$ is equal to (a) 1 (b) 7 (c) 11 (d) a value that cannot be computed without knowing the length of the vectors.	2	3	2

Part – B

SL. No	Questions	M	BT	CO
1.	Let u, v, w be three vectors in $\mathbb{R}^4$. You are told that $u \cdot v = u \cdot w$ Assert or Reject the following statement with proper justification: Since $u \cdot v = u \cdot w$, the vector $v = w$ vector.	4	2	1
2.	Find the eigenvalues and eigenvectors of the matrix $A = \begin{pmatrix}5 & 4\\4 & 5\end{pmatrix}$	6		1

3. Let W be the vector space spanned by the columns of A, where A is as given below. Find the Orthogonal Complement subspace of W.

$$A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \\ 0 & -1 \end{pmatrix}$$

6 | 3 | 1

4. Let

$$A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

Find the null space and column space of A. Specify the dimension of the null space of A and column space of A. What do you observe? Can you generalize this result for any square matrix?

10 | 3 | 2

5. For the matrix A given below, obtain the four fundamental subspaces associated with it.

$$A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \\ 0 & -1 \end{pmatrix}$$

You are expected to identify a suitable basis for each of the subspaces and the dimension for each of those subspaces. Tabulate your answer as follows:

Subspace	Basis	Dimension
Null space of A		
Column Space of A		
Null Space of A^T		
Column space of AT		

12 | 2 | 1

6. Given the following vectors, check if they are orthonormal or not.

$$\phi_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}, \phi_2 = \begin{pmatrix} 1 \\ -1 \\ 1 \\ -1 \end{pmatrix}, \phi_3 = \begin{pmatrix} 1 \\ 1 \\ -1 \\ -1 \end{pmatrix}, \phi_4 = \begin{pmatrix} 1 \\ -1 \\ -1 \\ 1 \end{pmatrix}$$

Express the vector

$$u = \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix}$$

as a linear combination of $\phi_1, \phi_2, \phi_3, \phi_4$.

12 | 3 | 2

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Date: 14.01.2026	**Improvement Test**	**Max. Marks: 50+10**
Semester: V	**UG**	**Duration: $1\frac{1}{2}$ Hrs + ½ Hr**
Title: Mathematical Algorithms for Artificial Intelligence		**Course Code: AI255BTC**

Part A

Q. No		PART A Quiz 1	M	BT	CO
1	a)	Which among the following can be the characteristic equation for a REAL SYMMETRIC MATRIX? a) $\lambda(\lambda^2+4)(\lambda+3)=0$ b) $\lambda(\lambda^4+16)=0$ c) $\lambda(\lambda^2-9)=0$ d) None of the above can be the characteristic equation for a real symmetric matrix	2	1	1
	b)	Let A be a 3x3 real matrix of rank 1. You are told that the trace of the matrix is 9. The eigenvalues of A are a) 0, 0, 3 b) 0, 0, -3 c) 0, 0, 9 d) 0, 0, -9	2	1	2
	c)	Given the matrix $A = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$, the pseudoinverse of this matrix is a) Non-existent for A, as it is a single column matrix b) $(A^TA)^{-1}A^T$ c) $A^T(AA^T)^{-1}$ d) A^{-1}	2	2	3
	d)	The eigenvalues of a projection matrix can only be a) ± 1 b) 0 and 1 c) 0 and -1 d) Determined based on the matrix.	2	2	2
	e)	**Statement 1:** Gram-Schmidt Orthonormalization always yields orthonormal vectors **Statement 2:** Orthonormal vectors are always linearly independent. a) Statement 1 and Statement 2 are ALWAYS TRUE b) Statement 1 and Statement 2 are ALWAYS FALSE c) Only Statement 1 is ALWAYS TRUE d) Only Statement 2 is ALWAYS TRUE	2	2	3

SL. No	Part B Questions	M	BT	CO
1.	Suppose A is an n x n orthogonal matrix, and $x \in \mathbb{R}^n$ prove or disprove that $\lVert Ax \rVert = \lVert x \rVert$, where $\lVert\ \rVert$ is the length of the vector. *No partial marks will be given if you take a specific numerical problem and solve.*	8	4	2
2.	Let u and v be two vectors in $\mathbb{R}^2$, such that $A = uv^T$. Obtain the SVD of the matrix A. *No partial marks will be given if you take a specific numerical problem and solve.*	8	3	2
3.	Given a singular value decomposition of an m x n matrix $A = U\Sigma V^T$, find an SVD of A^T. How are the singular values of A^T related to A?	8	3	2
4.	Show that if x is in both W and $W^{\perp}$, $x = 0$. *No partial marks will be given if you take a specific numerical problem and solve.*	8	4	2
5.	If P is a projection matrix, prove or disprove (I-P) is also a projection matrix, where I is the identity matrix. *Hint: Recall the definition of Projection matrix.*	8	4	2
6.	Let $M = I - 2uu^T$ where u is a unit vector in $\mathbb{R}^2$. This matrix M reflects every vector about the line passing through the origin and along the direction of **u**. Obtain the eigenvalues of the matrix M. OR (a) Prove that for a real symmetric matrix A, the eigenvectors corresponding to distinct eigenvalues are orthogonal to each other. (b) Obtain the conditions for solution to exist for the system of linear equations $Ax = b$. Assume A is a tall matrix with rank $r < n$, where n is the number of columns of A.	10	4	2